1T-1C DRAM CELL

with Cross-Coupled CMOS Sense Amplifier

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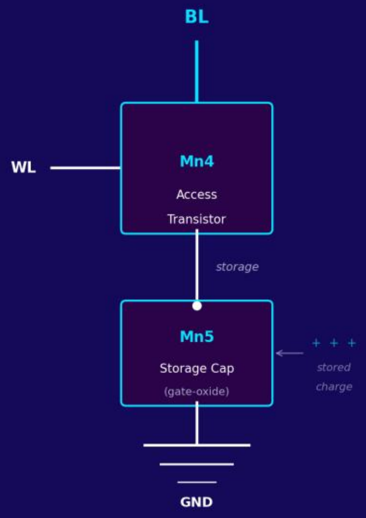
ECEN 4012

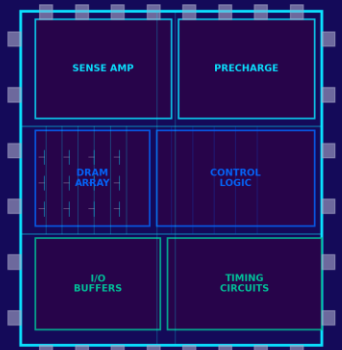
Siemens Tanner with Generic 250nm

WHAT IS DRAM?

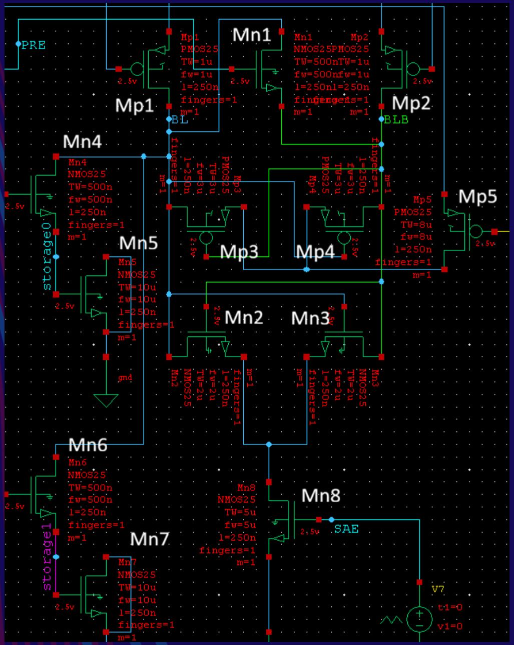
- Your laptop probably has 8 to 32 GB of it and it is the main memory in every computer
- Stores one bit as charge on gate-oxide capacitance
- "Dynamic" because that charge leaks. It must be refreshed constantly (~64 ms)
- Reading is destructive. It drains the capacitor
- The sense amplifier detects the charge and restores it



1 transistor + 1 capacitor = 1 bit



Generic 250nm | 2.5V | 13 transistors



WHAT WAS MADE

2-Cell DRAM Array

Mn4/Mn5 (Cell 0) + Mn6/Mn7 (Cell 1)
Shared BL/BLB bitlines

Cross-Coupled Sense Amp

Mp3/Mp4 (PMOS latch)
Mn2/Mn3 (NMOS latch)

Precharge Circuit

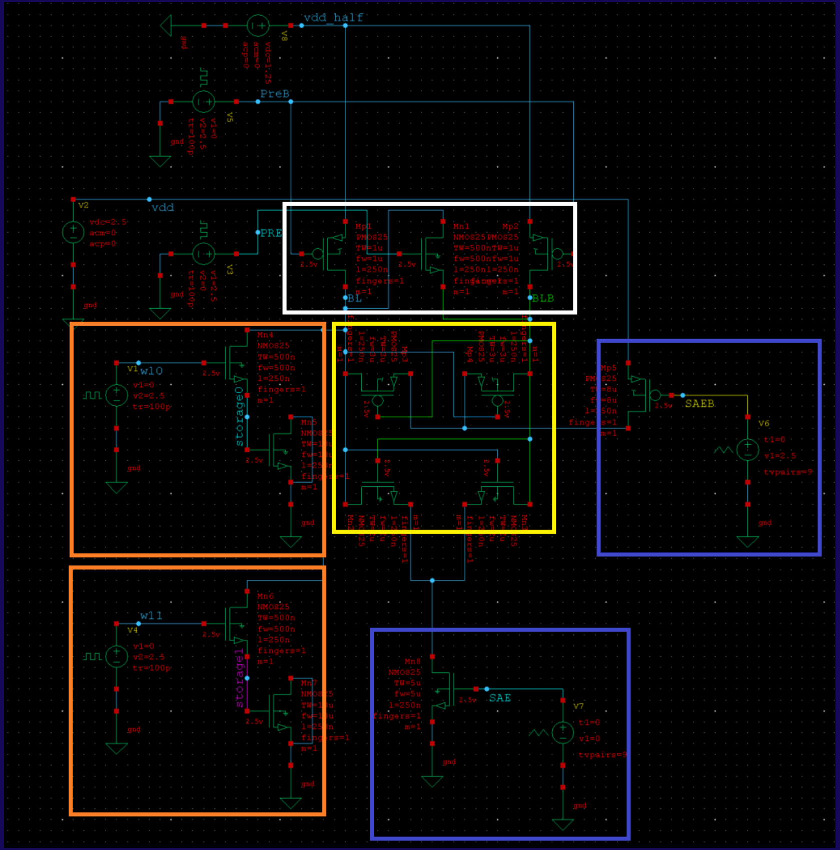
Mp1/Mp2 pull BL+BLB to VDD/2
Mn1 equalizer shorts both lines

SAE/SAEB Enable

Mp5 gates PMOS latch to VDD
Mn8 gates NMOS latch to GND

13 transistors using a Generic 250nm process with a 2.5V supply

COMPLETE SCHEMATIC



VDD/2 Supply
1.25V, precharge to mid-rail

PREB / PRE
Split precharge control signals

Mp1, Mn1, Mp2
Precharge + equalizer block

Mp3/Mp4 + Mn2/Mn3
Cross-coupled sense amp latch

Mp5 / Mn8
SAE/SAEB enable gating

Mn4/Mn5 + Mn6/Mn7
Cell 0 and Cell 1 (1T-1C each)

HOW A READ WORKS

PRECHARGE

0 to 8 ns

PRE goes high. Mp1 and Mp2 pull BL and BLB to 1.25V. Mn1 shorts both lines together so they equalize to the exact same voltage.

WORDLINE FIRES

10 ns

PRE falls. BL floats. WL asserts and connects the storage capacitor to BL. Charge sharing nudges BL slightly above or below 1.25V.

SENSE AMP ON

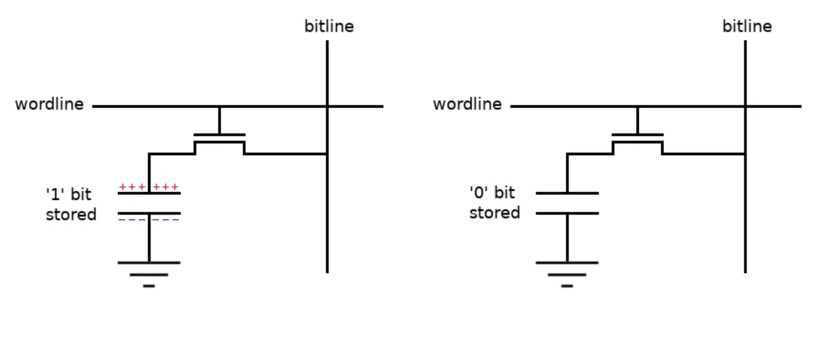
12 ns

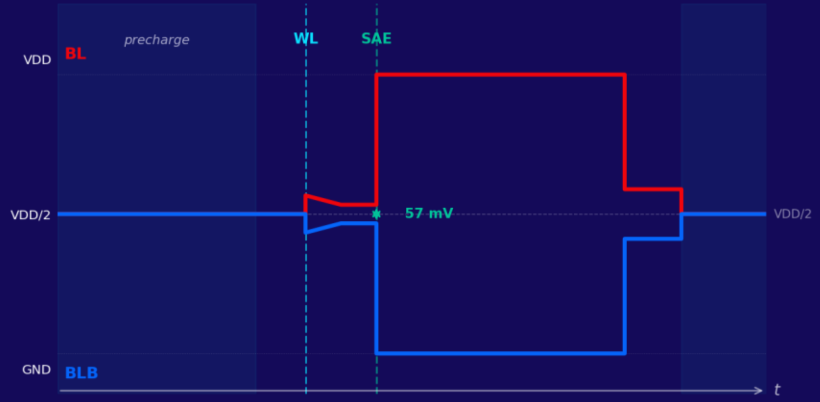
SAE fires 2ns later. The cross-coupled latch detects the tiny differential and amplifies it. BL snaps to VDD, BLB snaps to GND.

REFRESH

12 to 19 ns

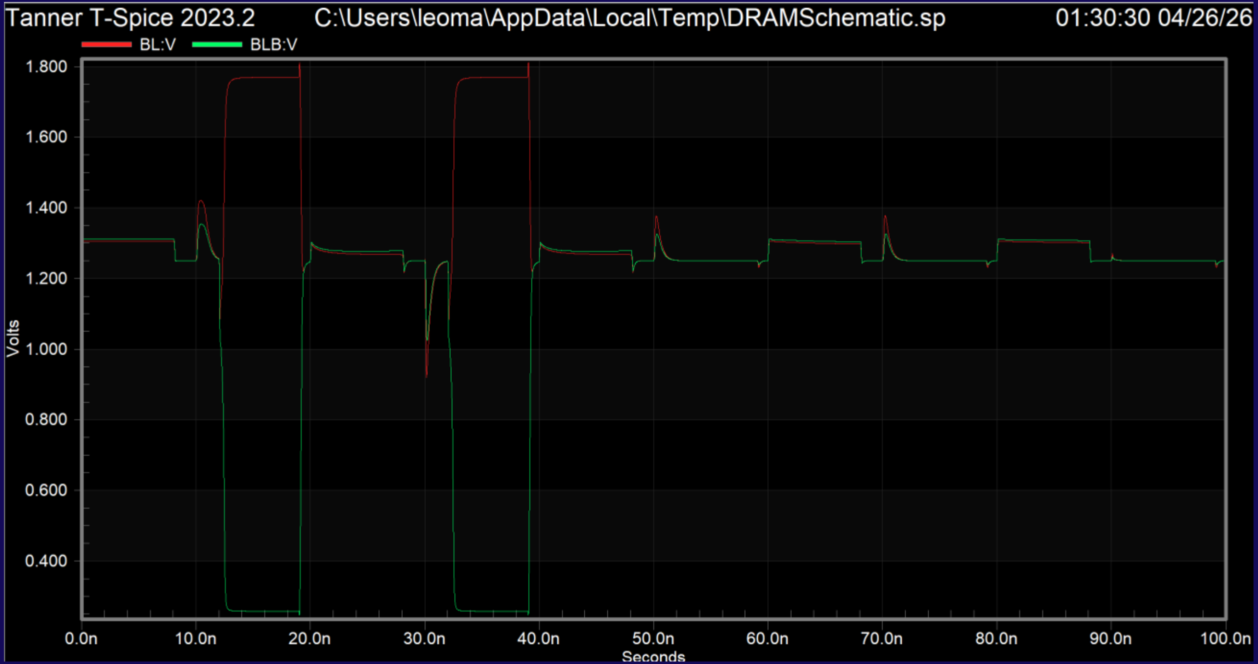
WL stays open while the sense amp drives BL. Current flows back through the access transistor and recharges the storage capacitor.





https://www.allaboutcircuits.com/technical-articles/introduction-to-dram-dynamic-random-access-memory/

SIMULATION: BITLINES



Cell 0 (t = 10 to 19ns)

BL rises -> reads logic 1

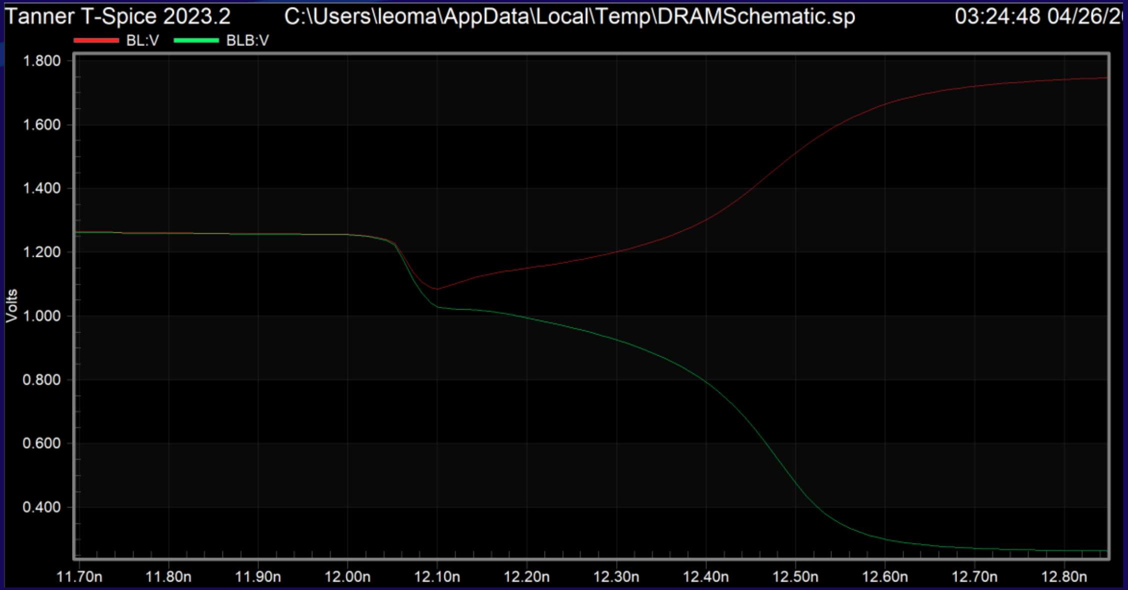
Cell 1 (t = 30 to 39ns)

BLB rises -> reads logic 0

Both cells confirmed

Opposite behavior

SENSE MARGIN: THE COOL PART



57 mV

Pre-latch differential

> 50 mV

Project target

~1.5 V

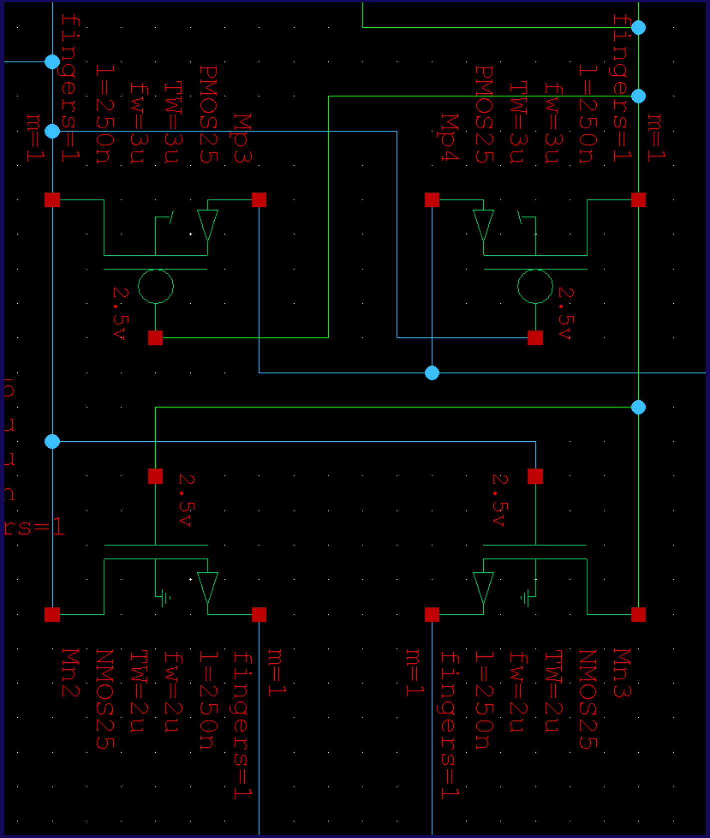
Post-latch swing

The sense amp detects a 57 mv signal and amplifies it to 1.5V

SOMETHING COOL: THE CROSS-COUPLED LATCH

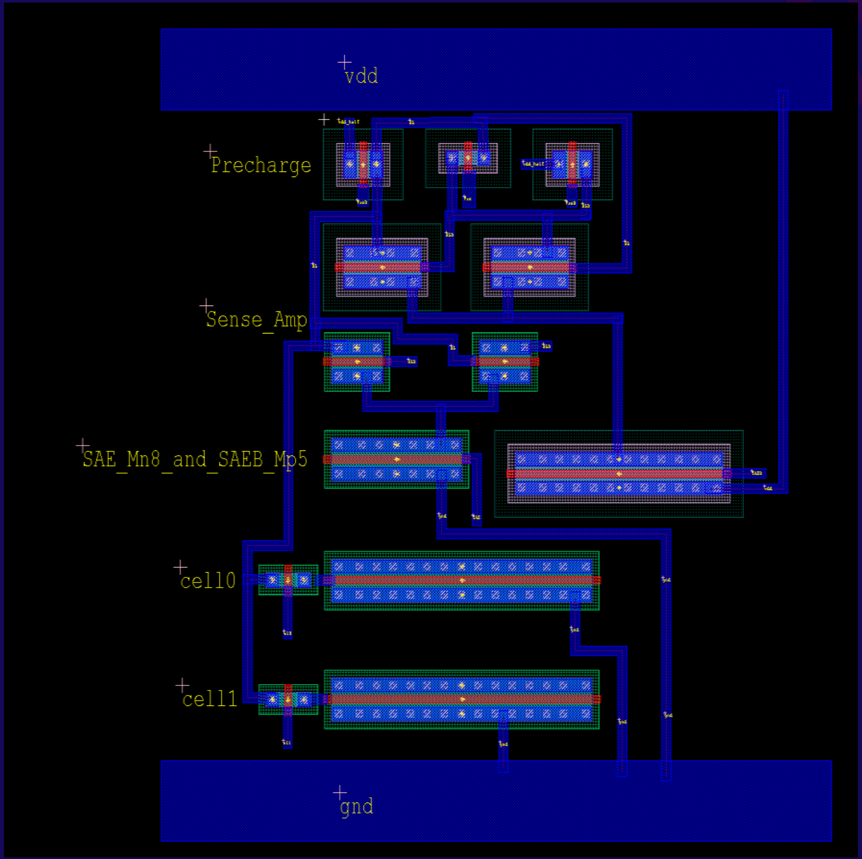
The sense amplifier is the most interesting circuit

- Two inverters wired back-to-back. Each output drives the other's input
- Stable until a very slight differential pushes it over to one side
- Any tiny nudge (57mV) causes it to snap to a definite state
- Positive feedback: the winner pulls the loser further down
- Goes from 57mV to 1.5V in about 300 picoseconds
- Sources are all timed.



This same topology is used in every SRAM cell, flip-flop, and register in a CPU.

PHYSICAL LAYOUT IN L-EDIT



Floorplan

- **VDD Rail (top)**
  2.5V power bus
- **Precharge Block**
  Mp1, Mn1, Mp2
- **Sense Amp PMOS**
  Mp3 / Mp4 latch
- **Sense Amp NMOS**
  Mn2 / Mn3 latch
- **SAE/SAEB Enable**
  Mn8 + Mp5
- **Cell 0**
  Mn4 + Mn5
- **Cell 1**
  Mn6 + Mn7
- **GND Rail (bottom)**
  Ground bus

RESULTS SUMMARY

57 mV
Sense Margin

> 50 mV
Target

1.5 V
Resolved swing

13
Transistors

Metric	Result	Status
Precharge to VDD/2	~1.3V (target 1.25V)	Pass
Cell 0: logic 1	BL high, BLB low	Pass
Cell 1: logic 0	BL low, BLB high	Pass
Sense margin	57 mV (target > 50 mV)	Pass
Storage node refresh	Partial (~1.7V restore)	Pass
Physical layout	13 transistors in L-Edit	Complete

TAKEAWAYS

DRAM stores one bit as charge. It leaks, so it must be refreshed constantly

The sense amp is a cross-coupled latch that turns a small 57mV signal into a much larger 1.5V

That same latch is in every SRAM cell, flip-flop, and register in your CPU

Questions?